

CMS and CMS Spread Options

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1 CMS Options

In this section we are mostly interested in analytical or semi-analytical methods to value CMS swaps. We describe two such methods we believe to be widely used in the market, the Replication and Hagan's method in [Hagan03a]. These two models share a lot of modelling concepts such that we start by describing the common points, before going into the specifics of each method.

We would like to stress the fact that the semi-analytical methods presented here both start from assuming the numeraire ratio (ZCB/annuity) to be a parametric function of the swap rate, which is an approximation of the reality where the yield curve has of course a term structure and the ratio depends on all the swap rates.

Once this assumption is made, the Replication is fully specified while Hagan's method requires further approximation, namely linearity of the numeraire ratio and log-normality of the swap rate. One can therefore view Hagan's method as an approximation of the Replication method, trading accuracy for the benefit of simplicity and speed.

1.1 Replication Method

The payoff of a CMS swaplet is

$$\text{Max}\left(w(S_{T_F} - K), 0\right) \quad (1)$$

where

- S_{T_F} is the swap rate fixed at T_F
- K is the strike
- $w = +1$ for a CMS caplet, $w = -1$ for a floorlet.

The cms option is paid at T_P and its present value is given by

$$V(w, K, T_F, T_P) = E^Q \left[e^{-\int_0^{T_P} r(u) du} \text{Max}\left(w(S_{T_F} - K), 0\right) \right] \quad (2)$$

where r is the discount rate. This can be valued by Monte-Carlo simulation after choosing a suitable term-structure model (such as Hull-White, LMM, etc...).

Alternatively one could attempt to value this payoff by implying its value from the knowledge of the swaption cube, which should be sufficient to determine the distribution of the future yield curve. In this document we choose to develop methodologies going in this direction.

The connection between CMS options and swaptions is the following. Let us work on the CMS caplet for shortness of notations, while extension to the floorlet is straightforward unless mentioned otherwise. After change of measure to the T_P -forward measure using (54), the CMS caplet PV has the expression

$$V_C(K, T_F, T_P) = P(0, T_P) E^{T_P} \left[(S_{T_F} - K)^+ \right] \quad (3)$$

On the other hand, a market payer swaption paying at T_F is calculated as

$$V_P(K, T_F) = A(0) E^A \left[(S_{T_F} - K)^+ \right]. \quad (4)$$

where A is the annuity.

The expressions (3) for the CMS caplet and (4) for a payer swaption are quite similar but evaluated in different measures. To connect the two prices we need to express them in the same measure, which we choose to be the annuity measure. We therefore change measure from T_P -forward to annuity in the expression of the CMS caplet.

A change of measure is performed by using the numeraire ratio

$$\alpha(t) = \frac{P(t, T_P)}{A(t)} \quad (5)$$

and the caplet price can be written as

$$V_C(K, T_F, T_P) = \frac{P(0, T_P)}{\alpha(0)} E^A \left[\alpha(T_F) (S_{T_F} - K)^+ \right] \quad (6)$$

where the numeraire ratio $\alpha(t)$ must be a Martingale in order for us to be able to change measure.

Now we can more easily realize the connection between CMS and swaptions, i.e. by comparing the caplet price (6) to the swaption price in (4). The link is provided by the Carr-Madan formula [CarrMadan99a], showing that any payoff that depends only on one underlying (at one time) can be written as a sum of call and put options on this underlying. More precisely, Carr-Madan formula reads

$$f(S) = f(x^*) + f'(x^*)(S - x^*) + \int_{-\infty}^{x^*} f''(x)(x - S)^+ dx + \int_{x^*}^{+\infty} f''(x)(S - x)^+ dx. \quad (7)$$

Using this formula with the swap rate as underlying and in the annuity measure, the call and put options become payer and receiver swaptions. This formula will thus allow us to write the CMS options as sums of swaptions, that is, provided we satisfy the assumptions under which the formula is valid. This point is where the core modelling assumption behind the replication methods lies.

Terminal Swap Rate Assumption (TSR)

Carr-Madan's formula is valid when the payoff function depends on only one underlying, so we will *assume that the numeraire ratio $\alpha(t)$ is a function of the swap rate S_t only*, i.e.

$$\alpha(t) = G(S_t). \quad (8)$$

Once a functional form $G(S)$ has been specified, the price is given by

$$V_C(K, T_F, T_P) = \frac{P(0, T_P)}{G(S_0)} E^A \left[G(S_{T_F}) (S_{T_F} - K)^+ \right] \quad (9)$$

and we apply Carr-Madan's formula in (7) to the function $f(S) = G(S)(S - K)^+$. After some algebra as described in appendix A.2, we find

$$\begin{aligned} \frac{V_C(K, T_F, T_P)}{P(0, T_P)} &= \frac{V_P(K)}{A(0)} + \frac{1}{G(S_0)A(0)} \left[(G(K) - G(S_0))V_P(K) + \int_K^{+\infty} (2G'(x) + G''(x)(x - K))V_P(x)dx \right] \\ \frac{V_F(K, T_F, T_P)}{P(0, T_P)} &= \frac{V_R(K)}{A(0)} + \frac{1}{G(S_0)A(0)} \left[(G(K) - G(S_0))V_R(K) - \int_{-\infty}^K (2G'(x) + G''(x)(x - K))V_R(x)dx \right] \\ \frac{V_S(T_F, T_P)}{P(0, T_P)} &= \frac{V_C(K, T_F, T_P)}{P(0, T_P)} - \frac{V_F(K, T_F, T_P)}{P(0, T_P)} + K \end{aligned} \quad (10)$$

where

- $V_C(K, T_F, T_P)$ is the CMS caplet
- $V_F(K, T_F, T_P)$ is the CMS floorlet
- $V_S(T_F, T_P)$ is the CMS swaplet (i.e. payoff S)

- $V_P(K, T_F, T_P)$ is the payer swaption
- $V_R(K, T_F, T_P)$ is the receiver swaption.

No-Arbitrage Conditions

For the absence of arbitrage to hold the change of measure between T_P -forward to annuity must be valid, which means that the numeraire ratio $\alpha(t)$ must be a Martingale in the annuity measure, that is

$$E^A[\alpha(T)] = \alpha(0). \quad (11)$$

Under the TSR assumption that underlies the replication methods, we have that

$$E^A[\alpha(T)] = E^A[G(S_T)] \quad (12)$$

$$\alpha(0) = G(S_0) \quad (13)$$

so that the no-arbitrage condition translates into

$$E^A[G(S_T)] \stackrel{?}{=} G(S_0). \quad (14)$$

This condition will not hold for generic choices of $G(S)$ (think of Jensen's inequality), but it can easily hold for the family of Linear Swap Rate models (LSR) which we focus on in this document, and describe below.

Additionally we mention the fact that the condition (14) needs further specifying. Indeed, so long as this condition is satisfied we are allowed to perform the change of measure. However, if the specified function $G(S_t)$ happens not to be equal to $P(t, T_P)/A(t)$, then we are no longer expanding on swaptions but on some other unknown product. As a consequence, we also need to impose this connection. We cannot do it exactly in general, but we should at least satisfy this condition on its firm moment, that is, in expectation. This leads to the additional constraint we would like to impose

$$E^A[G(S_T)] \stackrel{?}{=} G(S_0) \stackrel{?}{=} \frac{P(0, T_P)}{A(0)}. \quad (15)$$

Generic Replication

The Replication method can be applied as is, once the function $G(S)$ has been specified, simply by calculating the integrals in (10). Note that it is advised to satisfy the no-arbitrage conditions (14) and (15), which drastically limits the number of tractable models. In practice, all the models we are aware of assume linearity of the function $G(S)$. These are the so-called Linear Swap Rate models that we analyze below.

1.2 Linear Swap Rate Models

The family of LSR models assumes linearity of the function $G(S)$, which makes it easy to satisfy the no-arbitrage condition (14). Indeed, for any linear function $G(S) = G(S_0) + G'(S_0)(S - S_0)$ we have

$$\begin{aligned} E^A[G(S_T)] &= G(S_0) + G'(S_0)(E^A[S] - S_0) \\ &= G(S_0) \end{aligned} \quad (16)$$

since in the annuity measure $E^A[S] = S_0$. For further reference we define the ratio

$$R_0 = \frac{G'(S_0)}{G(S_0)}. \quad (17)$$

1.2.1 Formulation with integrals for generic distributions

Thanks to the linearity condition, we can rewrite the replication equations (10) as

$$\begin{aligned}
 \frac{V_C(K, T_F, T_P)}{P(0, T_P)} &= \frac{V_P(K)}{A(0)} + \frac{R_0}{A(0)} \left[(K - S_0) V_P(K) + 2 \int_K^{+\infty} V_P(x) dx \right] \\
 \frac{V_F(K, T_F, T_P)}{P(0, T_P)} &= \frac{V_R(K)}{A(0)} + \frac{R_0}{A(0)} \left[(K - S_0) V_R(K) - 2 \int_{-\infty}^K V_R(x) dx \right] \\
 \frac{V_S(T_F, T_P)}{P(0, T_P)} &= S_0 + 2 \frac{R_0}{A(0)} \left[\int_0^{+\infty} V_P(x) dx + \int_{-\infty}^0 V_R(x) dx - \frac{A(0)}{2} S_0^2 \right].
 \end{aligned} \tag{18}$$

where we have used the ratio R_0 defined in (17).

Note that to obtain this last equation on the CMS forward we used put-call parity and it applied it at ATM, but the same result can be obtained by applying Carr-Madan formula directly with $x^* = 0$. These expressions make it explicit that the prices can be calculated as a 'base price' ignoring the measure mismatch, plus convexity adjustments. The convexity adjustments are the second terms in the r.h.s.

Without further restricting assumptions on the distribution of the swap rate, we can simply calculate numerically the integrals in (18). The advantage of this approach is that it maximises the consistency with the market smile of the swaptions. The drawback is that numerical integrals must be calculated, with a slow down of the runtime compared to fully analytical approximations. Furthermore, it requires the choice of integration bounds and number of integration points, which can be a delicate issue.

1.2.2 Formulation with square swaptions

We are now going to give an equivalent form to these equations by noticing the identities

$$\frac{\partial \left[(S - x)^+ \right]^2}{\partial x} = -2(S - x)^+ \tag{19}$$

$$\frac{\partial \left[(x - S)^+ \right]^2}{\partial x} = 2(x - S)^+ \tag{20}$$

which lead to the appearance of the square swaption payoffs as

$$\int_K^{+\infty} (S - x)^+ dx = \frac{1}{2} \left[(S - K)^+ \right]^2 \tag{21}$$

$$\int_{-\infty}^K (x - S)^+ dx = \frac{1}{2} \left[(K - S)^+ \right]^2. \tag{22}$$

The forward option prices can then be rewritten as

$$\begin{aligned}
 \frac{V_C(K, T_F, T_P)}{P(0, T_P)} &= \frac{V_P(K)}{A(0)} + R_0 \left((K - S_0) \frac{V_P(K)}{A(0)} + E^A \left[\left((S - K)^+ \right)^2 \right] \right) \\
 \frac{V_F(K, T_F, T_P)}{P(0, T_P)} &= \frac{V_R(K)}{A(0)} + R_0 \left((K - S_0) \frac{V_R(K)}{A(0)} - E^A \left[\left((K - S)^+ \right)^2 \right] \right) \\
 \frac{V_S(T_F, T_P)}{P(0, T_P)} &= S_0 + R_0 \left(E^A \left[(S^+)^2 \right] + E^A \left[\left((-S)^+ \right)^2 \right] - S_0^2 \right).
 \end{aligned} \tag{23}$$

The advantage of this formulation is that if we choose a special dynamic for the swap rate so that square swaptions can be valued analytically, then the pricing of CMS options is fully analytical, i.e. does not need a numerical integration. We review two of these specific distributions below, the Log-Normal and Normal ones.

1.2.3 Log-Normal distribution

The choice of Log-Normal dynamic for S makes it possible to calculate the value of square swaptions exactly in analytical form. The square swaption closed-forms are calculated in appendix A.3.1. Defining $\omega = \pm 1$ for cap/floorlets, the expressions of the forward prices under a log-normal swap rate are

$$\frac{V_\omega(K, T_F, T_P)}{P(0, T_P)} = \frac{V_\omega(K)}{A(0)} + \omega R_0 S_0 \left(S_0 e^{\hat{\sigma}^2} \mathcal{N}(\omega d_{3/2}) - (S_0 + K) \mathcal{N}(\omega d_{1/2}) + K \mathcal{N}(\omega d_{-1/2}) \right) \quad (24)$$

$$\frac{V_S(T_F, T_P)}{P(0, T_P)} = S_0 + R_0 S_0^2 (e^{\hat{\sigma}^2} - 1) \quad (25)$$

where

$$\begin{aligned} \hat{\sigma} &= \sigma \sqrt{T} \\ d_\lambda &= \frac{\ln S_0 / K}{\hat{\sigma}} + \lambda \hat{\sigma}, \end{aligned} \quad (26)$$

$V_\omega(K)$ is the relevant swaption price (i.e. payer for caplets, receiver for floorlets), and σ is the swaption implied volatility. It is taken at strike K for the cap/floorlets and at ATM for the swaption.

The famous Hagan model in [Hagan03a] belongs to this family, with the specific choice of Swap Yield model for the function $G(S)$, see section 1.3.1.

1.2.4 Normal distribution

The exact closed-forms for the square swaptions under a Normal dynamic for the swap rate are calculated in appendix A.3.2. They lead to the following expressions for the CMS products

$$\frac{V_\omega(K, T_F, T_P)}{P(0, T_P)} = \frac{V_\omega(K)}{A(0)} + \omega R_0 \hat{\sigma}^2 \mathcal{N}\left(\omega \frac{S_0 - K}{\hat{\sigma}}\right) \quad (27)$$

$$\frac{V_S(T_F, T_P)}{P(0, T_P)} = S_0 + R_0 \hat{\sigma}^2 \quad (28)$$

$V_\omega(K)$ is the relevant swaption price, and σ is the swaption (Bachelier, i.e. Normal) implied volatility. It is taken at strike K for the cap/floorlets and at ATM for the swaption.

1.2.5 Shifted Log-Normal distribution

Denoting the shift by b , a shifted log-normal distribution for the spot S means that S is bounded from below by $-b$ and can be written as

$$S = -b + (S_0 + b) e^{-\frac{\hat{\sigma}^2}{2} + \hat{\sigma} u} \quad (29)$$

for some normally distributed random number u . The shifted spot $S + b$ is a log-normal variable, and option payoffs can be calculated with the usual Black-Scholes formula evaluated on the shifted forward $S_0 + b$ for the shifted strike $K + b$. We can thus rewrite the CMS prices as

$$\frac{V_\omega(K, T_F, T_P)}{P(0, T_P)} = \frac{V_\omega(K)}{A(0)} + \omega R_0 \tilde{S}_0 \left(\tilde{S}_0 e^{\hat{\sigma}^2} \mathcal{N}(\omega d_{3/2}) - (\tilde{S}_0 + \tilde{K}) \mathcal{N}(\omega d_{1/2}) + \tilde{K} \mathcal{N}(\omega d_{-1/2}) \right) \quad (30)$$

$$\frac{V_S(T_F, T_P)}{P(0, T_P)} = S_0 + R_0 \tilde{S}_0^2 (e^{\hat{\sigma}^2} - 1) \quad (31)$$

where

$$\begin{aligned}\tilde{S}_0 &= S_0 + b \\ \tilde{K} &= K + b\end{aligned}$$

and the d functions are evaluated on the ratio $\tilde{S}_0/\tilde{K}$.

1.3 Models in the Literature

All the models described below belong to the Linear Swap Rate family, i.e. those assuming that the function $G(S)$ is linear.

1.3.1 Linear Swap Yield Model

This family contains the famous Hagan model [Hagan03a] as well as the model by Mercurio et al. in [Mercurio05a]. The numeraire function $G(S)$ assume that all rates on the curve are the swap rate that is then used for discounting at all maturities in the annuity. Let us introduce the following notations:

- p is the number of cash-flows per year on the fixed side of the CMS index
- N is the total number of cash-flows on the fixed side of the CMS index
- $\Delta T = T_P - T_F$ is the delay between fixing and payment on the CMS swap (which may be different from that of the CMS index).

With the additional assumption of a regular schedule with year fraction $1/p$ on the fixed side, appendix A.4 shows that

$$G(S) = \frac{S}{\left(1 + \frac{S}{p}\right)^{p(T_P - T_F)} \left(1 - \frac{1}{\left(1 + \frac{S}{p}\right)^N}\right)}. \quad (32)$$

In the Mercurio-Pallavicini and Hagan models that we present under this family, this function is further linearized to

$$G(S) = G(S_0) + G'(S_0)(S - S_0). \quad (33)$$

The ratio $R_0 = \frac{G'(S_0)}{G(S_0)}$ is

$$R_0 = \frac{1}{S_0} \left(1 - \frac{S_0/p}{1 + S_0/p} \left[p\Delta T + \frac{N}{(1 + S_0/p)^N - 1}\right]\right) \quad (34)$$

and to match more closely typical notations in this family, we also define the function θ_0 as

$$\theta_0 = S_0 \frac{G'(S_0)}{G(S_0)} = 1 - \frac{S_0/p}{1 + S_0/p} \left[p\Delta T + \frac{N}{(1 + S_0/p)^N - 1}\right]. \quad (35)$$

As mentioned previously, one may now calculate the caplet, floorlet, and swaplet prices simply by numerical integration over the swaption prices. We review below what the two famous references [Hagan03a] and [Mercurio05a] bring on this subject.

Mercurio-Pallavicini

[Mercurio05a] follows precisely the above specification, but is only interested in displaying the result for the swaplets under the SABR model. We would like to stress the fact that the results above apply equally well to the caplets and floorlets, and to other models than SABR. Nevertheless, in order to make the connection of this model with the replication family clearer, we continue under these assumptions.

Since [Mercurio05a] assumes SABR, that is, a positive range for the swap and strikes, the forward swaplet price (a.k.a. the CMS adjusted forward) can be written as

$$\begin{aligned}\frac{V_S(T_F, T_P)}{P(0, T_P)} &= S_0 + 2 \frac{\theta_0}{S_0 A(0)} \left(\int_0^{+\infty} V_P(x) dx - \frac{A(0)}{2} S_0^2 \right) \\ &= S_0 + \theta_0 S_0 \left(\frac{2}{S_0^2} \int_0^{+\infty} E^A[(S - x)^+] dx - 1 \right)\end{aligned}\quad (36)$$

as the payer swaption has value $V_P(x) = A(0)E^A[(S - x)^+]$. Under SABR model (or any positive rate model), the expectation is $E^A[(S - x)^+] = \text{Black}(S_0, x, \sigma(S_0, x))$, i.e. Black formula for a call option with forward S_0 , strike K , and implied volatility $\sigma(S_0, K)$.

The integral over Black-Scholes formula is performed numerically. Note that without further difficulty, we could price caplets and floorlets in a very similar fashion.

Hagan

Hagan's method in [Hagan03a] makes the further simplifying assumption of *log-normality of the swap rate* S . As discussed in the previous section, the equivalent formulation in terms of square swaptions leads to an exact analytical form without need of integration. The resulting pricing formulas are (24) and (25) with the ratio R_0 in (34).

Extending to Shifted Log-Normal

As mentioned previously, a shifted log-normal price is simply obtained by evaluating Black-Scholes formula on the shifted forward at the shifted strike. With this rule we can readily apply the formulas (18) by using the shifted Black-Scholes (both in forward and strike) as swaption prices. Note that in principle, since the shifted log-normal distribution allows for negative strike options to have positive prices, the swaplet price should involve two integrations as the receiver part will have some values for negative strikes. Instead we can use the following strategy to have only one integration.

We go back to calculating the price of the swaplet by put-call parity, at the lower bound strike $-b$. Because this is the lower bound of the distribution, the integral on the receiver swaption vanishes. We obtain

$$\begin{aligned}\frac{V_S(T_F, T_P)}{P(0, T_P)} &= S_0 + R_0 \left[-(S_0 + b)^2 + 2 \int_{-b}^{+\infty} E^A[(S - x)^+] dx \right] \\ &= S_0 + \theta_0 \frac{(S_0 + b)^2}{S_0} \left[\frac{2}{(S_0 + b)^2} \int_{-b}^{+\infty} E^A[(S + b - (x + b))^+] dx - 1 \right] \\ &= S_0 + \theta_0 \frac{(S_0 + b)^2}{S_0} \left[\frac{2}{(S_0 + b)^2} \int_0^{+\infty} E^A[(S + b - x)^+] dx - 1 \right]\end{aligned}\quad (37)$$

where we have changed variable from x to $x + b$ in the last step. Since the process $S + b$ is log-normal, the price under the integral is $E^A[(S + b - x)^+] = \text{Black}(S_0 + b, x, \sigma(S_0 + b, x))$ i.e. it is Black-Scholes formula on the shifted forward at the strike x (not the shifted strike). Note that this formula is similar to but not exactly what we would have obtained by taking the shifted forward instead of S_0 in the convexity adjustment in (36). Also note that in the numerical integration for the caplets and floorlets, the range of integration must include negative rates going as low as $-b$ when relevant.

1.3.2 Hunt-Kennedy

This model was developed in Hunt-Kennedy (2000) and makes a specific assumption on the constant coefficient of the linear functional form. As the numeraire ratio function $G(S)$ is defined for the payment time T_P , we can write the most generic linear function $G(S)$ as

$$G(S) = b(T_P) + a(T_P)S. \quad (38)$$

Hunt-Kennedy makes the approximation that the coefficient $b(T_P)$ is independent of T_P , that is,

$$G(S) = b + a(T_P)S. \quad (39)$$

First we impose that the expectation of the numeraire ratio, i.e. that

$$\frac{P(0, T_P)}{A(0)} = E^A \left[\frac{P(T_F, T_P)}{A(T_F)} \right] = b + a(T_P)S_0 \quad (40)$$

which implies that

$$a(T_P) = \frac{\frac{P(0, T_P)}{A(0)} - b}{S_0}. \quad (41)$$

An additional constraint can be found by the identity

$$\sum_{i=1}^N \frac{P(T_F, T_P^i) \delta(T_F, T_P^i)}{A(T_F)} = 1 \quad (42)$$

which holds by definition of the annuity. Applying the numeraire ratio to it leads to

$$b \sum_{i=1}^N \delta(T_F, T_P^i) + S \sum_{i=1}^N \delta(T_F, T_P^i) a(T_P^i) = 1 \quad (43)$$

which must hold for any S . As a consequence, we find

$$b = \frac{1}{\sum_{i=1}^N \delta(T_F, T_P^i)} = \frac{p}{N} \quad (44)$$

where the last step holds when assuming regular intervals where p is the frequency. [Borst14a] remarks that this approximation should be all the more accurate as the rates are low (or equivalently that it will lose accuracy at high rates).

1.3.3 Mean Reversion

As with all LSR models, which take the generic form (38), we can use the no-arbitrage property which relates one parameter to the other through (41). While this equation gives the slope in terms of the constant, in the Mean Reversion model we start by finding the slope, by assuming an underlying 1-factor Normal mean-reverting model for the shape of the curve.

The coefficient $a(T_P)$ is the differential of the $G(S)$ function with respect to the swap rate. As we assume a 1-factor model with underlying variable r the short-rate, we use the chain rule to find $a(T_P)$ as

$$\begin{aligned} a(T_P) &= \frac{\partial G(S)}{\partial S} \\ &= \frac{\partial G[S(r)]}{\partial r} \left(\frac{\partial S(r)}{\partial r} \right)^{-1}. \end{aligned} \quad (45)$$

Then we assume the mean-reverting model form for the discount factors $P(t, T)$ as

$$P(t, T) = \frac{P(0, T)}{P(0, t)} C(t, T) e^{-B(t, T)r_t} \quad (46)$$

where the $C(t, T)$, $B(t, T)$ are deterministic functions of the initial curve and parameters, and in particular the $B(t, T)$ function (the only one whose actual expression we need), is

$$B(t, T) = \frac{1 - e^{-\kappa(T-t)}}{\kappa} \quad (47)$$

where κ is the mean reversion. The algebra is tedious but straightforward, leading to an expression of $a(T_P)$ in terms of the mean reversion κ and the initial yield curve (but not the mean-reverting model volatility). This calculation can be performed under the multiple curve framework. Once $a(T_P)$ is calculated, we obtain $b(T_P)$ by the no-arbitrage rule (41), i.e.

$$b(T_P) = \frac{P(0, T_P)}{A(0)} - a(T_P)S_0. \quad (48)$$

1.4 IR Server Models

1.4.1 CMS Forwards

The IR Server model for CMS forwards follows [Mercurio05a], was proposed in [MD-001] and was validated in [MV-032]. At this point in time, in order to avoid the potential moment explosion in the infinite integral in (36) due to the well-known problems in SABR's tails, the IR Server applied a threshold that acts as an upper bound on the swaption implied volatility. The threshold is an input chosen by the user.

The model was later extended in [MD-002] by following Antonov's closed-form in [Antonov12a] to correct the traditional SABR tails. This model was validated in [MV-040]. Currently both methods (Antonov and volatility threshold) are available, but due to practical considerations the volatility threshold method is preferred.

Multiple curves are handled by letting the forward swap rate S_0 be calculated exactly using different projection and discount curves. As noted in the validation report [MV-032], this is only a partial handling of the multiple curves as further corrections should in principle be included. However, [MV-032] concludes that the effect of the missing part is small.

The numerical integration is Gauss-Legendre with bounds as well as number of points specified by the user.

1.4.2 CMS Options

Two models are available for CMS options: LMM with Monte-Carlo simulation and Hagan's analytical form, see original [Hagan03a] and description in this document under section 1.3.1. Hagan's model has been described in the technical document [TD-001] and was validated in [MV-052].

2 CMS Spread Options

2.1 Survey of Literature

Surveying the literature on the subject, we can distinguish the following families of models

1. *Copula models*: these seem to be the most common. They assume that the marginal distributions of the swap rates have been obtained in a previous step that is pretty much the subject of section 1. Then they join the two marginal distributions using a copula. To do so they may or may not approximate the marginals in the first place. The copula may be Gaussian or more advanced.

These models have the practical advantages of speed, conceptual relative simplicity, and to apparently be very common. However they generally do not derive from a proper dynamics in the sense that the marginal distributions are usually obtained with Replication methods assuming a specific functional form of the numeraire ratio, an assumption that is difficult to justify on theoretical grounds.

Good reviews can be found in [Borst14a] and [Geissel12a], while the latter also has original developments.

2. *Term structure models*: i.e. using a full model of the yield curve with dynamics described as an SDE, etc... The most common choice in this family seems to be LMM, certainly due to the ability to model the correlation.

The advantages are that the model is fully consistent on theoretical grounds, and will typically be common with other IR derivatives. However, pricing is typically done through Monte-Carlo which is very slow. Some authors have been able to obtain approximate closed-forms.

See [Lutz09a] for an example.

3. *Multi-variate models*: they postulate a two-dimensional dynamics for the underlyings and project it onto a one-dimensional dynamics for which closed-forms may exist.

See [Kienitz10a] for an example, and [Borst14a] for a review.

4. *Mapping models*: survey in progress. These seem to relax the core simplifying assumption behind the Replication models into allowing a dependency of the numeraire ratio on all swap rates. They seem allow for the pricing of CMS spread options, but it is not clear to us at the moment how these models work.

See [Piterbarg12b].

2.2 Copula Models

2.2.1 Principles

Overview

Broadly speaking the copula models for CMS spread pricing work in the following way:

1. Obtain the (marginal) distribution of each swap rate
2. Choose a copula to describe their joint distribution
3. Integrate (2D) the joint distribution multiplied by the payoff to obtain the price.

Subtleties and difficulties

Each point above has its difficulties.

1. *Marginal distributions*: these are not easy to obtain. They are typically known only indirectly through pricing of the CMS options, which is itself a complex problem. Note that with the view to joining the distributions with the copula, the two distributions of the swap rates must be expressed in the same measure (typically the T_P -forward where T_P is the payment time), while swap rate distributions are most easily expressed in their respective annuity measures. Therefore a change of measure has to be applied in order to bring the two distributions in the same measure.

Moreover, obtaining well-defined (i.e. positive, summing to 1) distributions is not an easy task. The classic SABR model, for instance, is known to lead to ill-defined distributions in the tails. The Benaim extrapolation of it, while resolving the issue in most situations, is not guaranteed to lead to positivity at every point. Free-boundary SABR is numerically heavy and suffers from difficult numerical treatment at a number of points, etc...

Finally, in order to reach practical pricing of the CMS spread payoff, we will often need to work on specific types of distributions that lead to tractable integration. There is only a restricted number of such distributions, such that a projection or fit of the true distribution onto a specific, more tractable one, may have to be performed before reaching the copula step.

2. *Choice of copula*: not many choices of copulas lead to tractable pricing while at the same time having realistic and desirable properties.
3. *Integration of the payoff*: typically a 2-dimensional integration. 2-dimensional integrations are generally tedious and not very practical numerical processes. Most practical models will make specific choices of marginal and/or joint distributions to allow for at least partial analyticity, reducing the integration to at worse a 1-dimensional numerical one. While this makes pricing more practical, it still leaves the typical difficulty of numerical integration, which are the choices of integration bounds and the trade-off between accuracy and speed in terms of number of sampling points.

2.2.2 Marginal Distributions

We consider two possible strategies here:

1. *Direct distributions*: assuming that the choice of copula is such that we will effectively integrate the copula together with the marginal density, we strive to obtain the marginal distributions themselves. Each swap rate's distribution is typically known in that swap rate's annuity measure (thanks to the swaption market). We can either perform a change of measure to obtain the distribution in the T_P -forward measure, or calculate the CMS cap/floorlet/forward prices and derive the T_P -forward distribution from them.

Note that the process of deriving a probability density out of prices requires the calculation of the second differential of the prices, which is not always an easy and robust task.

2. *Through prices*: if the final integration (of the CMS spread payoff) can be done analytically or semi-analytically thanks to a specific choice of marginal and copula models, the specific marginal models will have parameters that need to be fit to the CMS option data. This can typically be done by fitting the prices of the marginal model to the prices of the CMS option model.

The connection between distribution and prices is given by the identity

$$\frac{\partial^2 E^N[(S - K)^+]}{\partial K^2} = p^N(K) \quad (49)$$

where $E^N[\cdot]$ is the expectation in the N -measure and $p^N(\cdot)$ is the probability density in the same measure. This is proved by differentiating twice w.r.t. the strike inside the expectation. Note that the probability density is tied to a specific measure, the one that has been used to calculate the prices.

Since we will be looking for the marginal distributions under the T_P -forward measures, we will need to derive them from the prices of CMS options since these are precisely call/put payoffs in that measure (by opposition to swaptions that are call/put payoffs in the annuity measure).

2.2.3 Marginal Model: Gaussian

The Normal (Gaussian) distribution is specified by its mean and volatility. As explained in the previous section, we are working in the T_P -forward measure, which implies that the mean is the CMS forward rate (not the forward swap rate). Under a Normal distribution, the option price follows Bachelier formula with the distribution mean as forward and the distribution volatility as implied volatility. We propose to use the ATM CMS caplet to imply the distribution volatility, through inversion of Bachelier formula (with the CMS forward rate as forward).

2.2.4 Marginal Model: Skewed Student-t

The Student-t distribution (and its skewed generalization) is often used in the context of Risk models but rarely in pricing. [Geissel12a] proposes to do so for the marginal distribution of the swap rates. We devote appendix A.6 to the description of this distribution.

2.3 IR Server Models

ToDo

3 Documentation Sign Off

Date	Version	Author	Change Details
05/04/2017	0.0	Sebastien Gurrieri	Initial version
10/04/2017	0.1	Sebastien Gurrieri	Adding a few numeraire models, discrete calculation of weights, references

A Toolkit

A.1 Change Of Measure

A.1.1 Generic formulation with two numeraires

Changing between numeraires N_1 and N_2

$$E_t^{N_1}[\frac{N_1(t)}{N_1(T)} R] = E_t^{N_2}[\frac{N_2(t)}{N_2(T)} R] \quad (50)$$

$$E_t^{N_1}[R] = E_t^{N_2}[\frac{N_1(T)N_2(t)}{N_1(t)N_2(T)} R] \quad (51)$$

or equivalently using the likelihood process $M(t)$

$$E_t^{N_1}[R] = E_t^{N_2}[\frac{M(T)}{M(t)} R]. \quad (52)$$

Then the likelihood process is a Martingale in N_2 since

$$E_t^{N_2}[M(T)] = M(t). \quad (53)$$

A.1.2 MMA and change to T -forward

By taking $N_1(t) = P(t, T)$ and $N_2(t) = e^{\int_0^t c(u) du}$ we obtain the change of measure from T -forward to collateral

$$E_t^T[R] = E_t^Q[\frac{e^{-\int_t^T c(u) du}}{P(t, T)} R]. \quad (54)$$

The likelihood process can be defined as

$$M(t) = \frac{P(t, T)e^{-\int_0^t c(u) du}}{P(0, T)} \quad (55)$$

and can be checked to satisfy the change of measure formula as

$$E_t^T[R] = E_t^Q[\frac{M(T)}{M(t)} R] \quad (56)$$

with the specific form of $M(\cdot)$ defined in (55). We can then invert to get the change from collateral to T -forward measure as

$$E_t^Q[R] = E_t^T[e^{\int_t^T c(u) du} P(t, T) R]. \quad (57)$$

A.2 Applying Carr-Madan

Our starting point is the pricing formula of a CMS caplet once the functional form of the numeraire ratio has been specified, such that the caplet price is

$$\begin{aligned}
V_C(K, T_F, T_P) &= A(0)E^{A_{T_F}} \left[f(S_{T_F}) \right] \\
f(S) &= G(S)(S - K)^+
\end{aligned} \tag{58}$$

where $G(S)$ is the functional form of the numeraire ratio in the specified model. In order to apply the Carr-Madan formula we need the differentials of $f(S)$, which we calculate below

$$\begin{aligned}
f(S) &= G(S)(S - K)^+ \\
f'(S) &= G(S)H(S - K) + G'(S)(S - K)^+ \\
f''(S) &= G(S)\delta(S - K) + 2G'(S)H(S - K) + G''(S)(S - K)^+
\end{aligned}$$

where $H(\cdot)$ and $\delta(\cdot)$ are the Heaviside and Dirac functions. Now we can calculate the caplet price by apply Carr-Madan on a central point $x^* < K$. Thanks to this choice, we can say that

$$\begin{aligned}
f(x^*) &= f'(x^*) = 0 \\
f''(x) &= 0 \quad \forall x \leq x^*
\end{aligned}$$

which leads us to

$$\begin{aligned}
V_C(K, T_F, T_P) &= A(0) \int_{x^*}^{+\infty} \left(G(x)\delta(x - K) + 2G'(x)H(x - K) + G''(x)(x - K)^+ \right) E^{A_{T_F}} \left[(S_{T_F} - x)^+ \right] dx \\
&= G(K)V_P(K) + \int_K^{+\infty} \left(2G'(x) + G''(x)(x - K) \right) V_P(x) dx
\end{aligned} \tag{59}$$

where $V_P(x)$ is the payer swaption PV for strike x .

The calculation for floorlet follows analogously. The payoff function $h(S)$ satisfies

$$\begin{aligned}
h(S) &= G(S)(K - S)^+ \\
h'(S) &= G(S)(H(S - K) - 1) + G'(S)(K - S)^+ \\
h''(S) &= G(S)\delta(S - K) + 2G'(S)(H(S - K) - 1) + G''(S)(K - S)^+
\end{aligned}$$

and by taking $x^* > K$ we find the floorlet price

$$V_F(K, T_F, T_P) = G(K)V_R(K) + \int_{-\infty}^K \left(-2G'(x) + G''(x)(K - x) \right) V_R(x) dx \tag{60}$$

where $V_R(x)$ is the receiver swaption PV for strike x . The CMS swaplet (payoff $S - K$) can be obtained by put-call parity as

$$V_S(K, T_F, T_P) = V_C(K, T_F, T_P) - V_F(K, T_F, T_P). \tag{61}$$

A.3 Square Options

A.3.1 Square Option of a Log-Normal Variable

Here we calculate the expectation of the square option payoff

$$E \left[\left((X - K)^+ \right)^2 \right] \tag{62}$$

where the underlying variable X is lognormal with volatility σ and mean X_0 in the measure behind the expectation operator $E[\cdot]$. We write this variable as

$$X = X_0 e^{-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}u} \quad (63)$$

for a normally distributed u , and simplify the notations as

$$\begin{aligned} X &= \hat{X}_0 e^{\hat{\sigma}u} \\ \hat{\sigma} &= \sigma\sqrt{T} \\ \hat{X}_0 &= X_0 e^{-\frac{1}{2}\hat{\sigma}^2}. \end{aligned}$$

The payoff can then be calculated as

$$\begin{aligned} E\left[\left((X - K)^+\right)^2\right] &= \frac{1}{\sqrt{2\pi}} \int_{-d_{-1/2}}^{+\infty} (\hat{X}_0 e^{\hat{\sigma}u} - K)^2 e^{-\frac{u^2}{2}} du \\ &= \frac{1}{\sqrt{2\pi}} \int_{-d_{-1/2}}^{+\infty} (\hat{X}_0^2 e^{2\hat{\sigma}u} - 2\hat{X}_0 K e^{\hat{\sigma}u} + K^2) e^{-\frac{u^2}{2}} du \end{aligned} \quad (64)$$

where

$$d_\lambda = \frac{\ln X_0/K}{\hat{\sigma}} + \lambda \hat{\sigma}. \quad (65)$$

Next we take the following property

$$\int_x^{+\infty} e^{\alpha\sigma u} e^{-\frac{u^2}{2}} du = e^{\frac{1}{2}\alpha^2\sigma^2} \sqrt{2\pi} \mathcal{N}(-x + \alpha\sigma) \quad (66)$$

and apply it to the 3 terms in the expectation, finding

$$\begin{aligned} E\left[\left((X - K)^+\right)^2\right] &= \frac{1}{\sqrt{2\pi}} \int_{-d_{-1/2}}^{+\infty} (\hat{X}_0^2 e^{2\hat{\sigma}u} - 2\hat{X}_0 K e^{\hat{\sigma}u} + K^2) e^{-\frac{u^2}{2}} du \\ &= \hat{X}_0^2 e^{\hat{\sigma}^2} \mathcal{N}(d_{3/2}) - 2\hat{X}_0 K \mathcal{N}(d_{1/2}) + K^2 \mathcal{N}(d_{-1/2}). \end{aligned} \quad (67)$$

A.3.2 Square Option of a Normal Variable

Now we assume the Normal dynamic for the variable X , that is, we write X as

$$X = X_0 + \hat{\sigma}u \quad (68)$$

for a normally distributed u with standard deviation $\hat{\sigma} = \sigma\sqrt{T}$. Then we find

$$E\left[\left((X - K)^+\right)^2\right] = \hat{\sigma}^2 \left[(1 + H^2) \mathcal{N}(-H) - \frac{H}{\sqrt{2\pi}} e^{-\frac{H^2}{2}} \right] \quad (69)$$

$$E\left[\left((K - X)^+\right)^2\right] = \hat{\sigma}^2 \left[(1 + H^2) \mathcal{N}(H) + \frac{H}{\sqrt{2\pi}} e^{-\frac{H^2}{2}} \right] \quad (70)$$

$$H = \frac{K - X_0}{\hat{\sigma}}. \quad (71)$$

A.4 More Details on the Swap Yield Model

Under the notations and assumptions of the Swap Yield Model as described in section 1.3.1, we can calculate that

$$P(t, T_P) = P(t, T_F) \frac{P(t, T_P)}{P(t, T_F)} = \frac{P(t, T_F)}{(1 + \frac{S}{p})^{p(T_P - T_F)}} \quad (72)$$

$$A(t) = P(t, T_F) \sum_{i=1}^N \frac{1}{p} \frac{1}{(1 + \frac{S}{p})^i} = \frac{P(t, T_F)}{S} \left(1 - \frac{1}{(1 + \frac{S}{p})^N} \right) \quad (73)$$

so that the numeraire ratio function is

$$G(S) = \frac{S}{(1 + \frac{S}{p})^{p(T_P - T_F)} \left(1 - \frac{1}{(1 + \frac{S}{p})^N} \right)}. \quad (74)$$

Note that at $S = 0$ this function takes the well-defined value

$$G(0) = \frac{p}{N}. \quad (75)$$

Let us first define the terms

$$\begin{aligned} D &= 1 + \frac{S}{p} \\ \Delta T &= T_P - T_F. \end{aligned}$$

We can then write the function $G(S)$ as

$$G(S) = \frac{S}{D^{p\Delta T} (1 - D^{-N})}. \quad (76)$$

The first differential is then

$$G'(S) = \frac{D^{N-p\Delta T-1}}{(D^N - 1)^2} \left((D - S\Delta T)(D^N - 1) - \frac{N}{p} S \right) \quad (77)$$

and we find the function θ as

$$\theta = S \frac{G'(S)}{G(S)} = 1 - \frac{S/p}{1 + S/p} \left[p\Delta T + \frac{N}{D^N - 1} \right]. \quad (78)$$

A.5 Discrete Weights in Replication

In this section we describe an explicit strategy to calculate the weights in the Replication expansions (10) in the case of CMS caplets and floorlets. The idea is to shortcut, or more precisely, adapt the calculation of the differentials of the function $G(S)$ to a specific strike grid and knowing the specific payoff. We start with the caplets to illustrate the principles.

We write the "payoff" as $G(S)H(S)$ to be more generic, in the sense that the conclusions apply to any payoff function $H(S)$ that is non-zero only for $S > K$. The Replication formula tells us that this payoff can be written only in terms of payer swaptions, so we write

$$G(S)H(S) = \sum_{i=1}^N w_i (S - K_i)^+ \quad (79)$$

where we have chosen a grid of N strikes $\{K\}_{i=1..N}$, and where $K_1 = K$. In the below we obtain a few properties that lead us to the values of the weights w_i .

Property 1.

$$G(K_{j+1})H(K_{j+1}) = \sum_{i=1}^j w_i (K_{j+1} - K_i)^+ \quad (80)$$

Proof. This is the result of applying the payoff at the strike point K_{j+1} and truncating the sum to the only non-zero terms, due to the fact that $(K_{j+1} - K_i)^+ = 0$ when $i \geq j + 1$. $\square$

Property 2.

$$\sum_{i=1}^j w_i = \frac{G(K_{j+1})H(K_{j+1}) - G(K_j)H(K_j)}{K_{j+1} - K_j} \quad (81)$$

Proof. First we use the following property

$$(K_{j+1} - K_i)^+ = (K_j - K_i)^+ + (K_{j+1} - K_j)^+, \quad i \leq j \quad (82)$$

which is straightforward when looking separately at the cases $i = j$ and $i < j$, and considering the ordered structure of the strike grid. Using this identity in property (1) leads to

$$G(K_{j+1})H(K_{j+1}) = G(K_j)H(K_j) + (K_{j+1} - K_j) \sum_{i=1}^j w_i \quad (83)$$

which gives the sum of the weights. $\square$

Property 3.

$$\begin{aligned} w_{j+1} &= \frac{G(K_{j+2})H(K_{j+2}) - G(K_{j+1})H(K_{j+1})}{K_{j+2} - K_{j+1}} - \frac{G(K_{j+1})H(K_{j+1}) - G(K_j)H(K_j)}{K_{j+1} - K_j} \\ w_1 &= G(K_2). \end{aligned} \quad (84)$$

Proof. The first line comes from calculating the difference $\sum_{i=1}^{j+1} w_i - \sum_{i=1}^j w_i$ and using property 2. The second line comes from using property 2 for $j = 1$, remembering that $K_1 = K$. $\square$

A.6 Skewed Student-t Distribution

We start by summarizing the definition and properties of the original and most basic Student-t distribution. Then we describe a general and very simple process (hybridizing) to obtain more advanced distributions (asymmetric, fat tails, etc...) out of simple ones, while retaining good tractability properties. Finally we apply it to the Student-t distribution to generate the skewed Student-t chosen by [Geissel12a].

A.6.1 Student-t Distribution

The Student-t is a fat-tailed and symmetric parametric distribution. It has 3 parameters: μ , akin to a location or mean, σ , akin to a volatility, and d , the number of degrees of freedom which parameterizes the fatness of the tails.

The standard Student-t density is

$$f_d(x) = \frac{\Gamma(\frac{d+1}{2})}{\Gamma(\frac{d}{2})\sqrt{d\pi}} \left(1 + \frac{x^2}{d}\right)^{-\frac{d+1}{2}}. \quad (85)$$

The full Student-t density $f_{\mu,\sigma,d}$ is obtained by centring and scaling as

$$f_{\mu,\sigma,d}(x) = \frac{1}{\sigma} f_d\left(\frac{x-\mu}{\sigma}\right). \quad (86)$$

We denote by $F_d(x)$ ¹ the CDF of the standard Student-t in (85). Then the CDF of the full Student-t is

$$F_{\mu,\sigma,d}(x) = F_d\left(\frac{x-\mu}{\sigma}\right). \quad (87)$$

We are not aware of an analytical form for the inverse CDF so we calculate it by numerical solving. Assuming we are able to calculate this inverse numerically and denote it by $F_d^{-1}(y)$ for the standard Student-t, we can then write the inverse of the centred Student-t as

$$F_{\mu,\sigma,d}^{-1}(y) = \mu + \sigma F_d^{-1}(y). \quad (88)$$

A.6.2 Hybrid Distributions

A hybrid distribution PDF $f(x)$ is obtained from two distributions with PDFs $f_1(x)$ and $f_2(x)$ as

$$f(x) = a_1 f_1(x) \mathbf{1}_{x < x_0} + a_2 f_2(x) \mathbf{1}_{x \geq x_0} \quad (89)$$

where $0 \leq a_1, a_2 \leq 1$ are constant coefficients and x_0 is a junction point. The coefficients are found so that the density $f(x)$ is a proper probability measure, that is, we impose

$$\begin{aligned} f(x_0^+) &= f(x_0^-) \\ \int_{-\infty}^{+\infty} f(x) dx &= 1 \end{aligned}$$

where $x_0^\pm$ means limits below(-) and above(+) x_0 . These consistency conditions translate into

$$\begin{aligned} a_1 f_1(x_0) &= a_2 f_2(x_0) \\ a_1 F_1(x_0) + a_2 (1 - F_2(x_0)) &= 1 \end{aligned}$$

which leads to the solution

$$\begin{aligned} a_1 &= \frac{1}{F_1(x_0) + (1 - F_2(x_0)) \frac{f_1(x_0)}{f_2(x_0)}} \\ a_2 &= a_1 \frac{f_1(x_0)}{f_2(x_0)}. \end{aligned}$$

The CDF of the hybrid is

$$\begin{aligned} F(x) &= a_1 F_1(x), \quad x < x_0 \\ F(x) &= a_1 F_1(x_0) + a_2 (F_2(x) - F_2(x_0)), \quad x \geq x_0 \end{aligned} \quad (90)$$

while its CDF inverse is

$$\begin{aligned} F^{-1}(y) &= F_1^{-1}\left(\frac{y}{a_1}\right), \quad 0 < y \leq a_1 F_1(x_0) \\ F^{-1}(y) &= F_2^{-1}\left(\frac{y - 1 + a_2}{a_2}\right), \quad a_1 F_1(x_0) \leq y \leq 1. \end{aligned} \quad (91)$$

¹Can be calculated using the incomplete beta functions.

A.6.3 Hybridizing the Student-t Distribution

There are several ways to create a distribution with similar tail properties as the Student-t but in a non-symmetric way. Most methods tend to be rather complex and computationally intensive, while we think that the hybridization above is a much simpler and efficient way of achieving the result. It is also the method chosen by [Geissel12a].

Hybridizing two Student-t following the process in section A.6.2 above leads to a distribution with more flexibility, and in particular with control over the two tails independently while the original Student-t has the same tail fatness on both sides.

We could in principle make use of the full freedom of the hybridization, that is, choose the two distributions to have different centre, scale and degrees of freedom, and take the junction point to be some value that would be an additional parameter, so that we have a maximum number of parameters of 7. Instead [Geissel12a] reduces the freedom by constraining some of the parameters, reducing their number to 4, which incidentally is the number of SABR parameters.

We keep as a note for later that Geisselmeyer's reduction is not the only possibility, others can easily be imagined and used in case of numerical difficulties during testing. Moreover, we may want to keep some of the freedom of the full hybrid in situations where the target model is not SABR but another model with more parameters, such as Benaim's SABR or ZABR.

Geisselmeyer's parameterization

[Geissel12a] reduces the freedom of the hybrid distribution in the following manner:

- the two distributions have the same centres and degrees of freedom
- the junction point is this common centre.

[Geissel12a] further introduces the skew parameter ϵ . Replacing his skewed-t in the broader context of hybrid distributions, we can see that his parameterization amounts to the hybridization of two Student-t distributions with the following parameters

$$\begin{aligned} f_1(x) &= f_{\mu, \frac{\sigma}{\epsilon}, d} \\ f_2(x) &= f_{\mu, \epsilon\sigma, d}. \end{aligned}$$

From this we can check that the hybrid distribution formulas lead to the formula in [Geissel12a] for the probability density. Indeed, our coefficients a_1, a_2 for the hybrid distribution are solved as

$$\begin{aligned} a_1 &= \frac{2}{1 + \epsilon^2} \\ a_2 &= \frac{2\epsilon^2}{1 + \epsilon^2} \end{aligned}$$

leading to

$$f_{\mu, \sigma, \epsilon, d}(x) = \frac{2\epsilon}{\sigma(1 + \epsilon^2)} f_d\left(\frac{\epsilon(x - \mu)}{\sigma}\right) \mathbf{1}_{x < \mu} + \frac{2\epsilon}{\sigma(1 + \epsilon^2)} f_d\left(\frac{x - \mu}{\epsilon\sigma}\right) \mathbf{1}_{x \geq \mu}. \quad (92)$$

However, our result on the CDF is different from that of [Geissel12a]. We find

$$F_{\mu, \sigma, \epsilon, d}(x) = \frac{2}{1 + \epsilon^2} F_d\left(\frac{\epsilon(x - \mu)}{\sigma}\right), \quad x < \mu \quad (93)$$

$$F_{\mu, \sigma, \epsilon, d}(x) = \frac{1 - \epsilon^2}{1 + \epsilon^2} + \frac{2\epsilon^2}{1 + \epsilon^2} F_d\left(\frac{x - \mu}{\epsilon\sigma}\right), \quad x \geq \mu. \quad (94)$$

We think that Geisselmeyer's formula is surprising as it does not appear to be continuous at the centre point μ for values $\epsilon \neq 1$. Finally our inverse CDF is

$$F_{\mu,\sigma,\epsilon,d}^{-1}(y) = \mu + \frac{\sigma}{\epsilon} F_d^{-1}\left(\frac{y(1+\epsilon^2)}{2}\right) \mathbf{1}_{y < \frac{1}{1+\epsilon^2}} - \sigma \epsilon F_d^{-1}\left(\frac{(1-y)(1+\frac{1}{\epsilon^2})}{2}\right) \mathbf{1}_{y \geq \frac{1}{1+\epsilon^2}}. \quad (95)$$

Note that we are in agreement with [Geissel12a] on the inverse.

A.6.4 Skewed Student-t Option Price

Property 4. *The CMS caplet under the skewed student-t distribution as defined in (92) is*

$$V_C = \begin{cases} \mu - K + 2\sigma(\epsilon - \frac{1}{\epsilon}) \frac{d}{d-1} f_d(0) + \frac{2\sigma}{\epsilon(1+\epsilon^2)} \left[\frac{d+\beta^2}{d-1} f_d(\beta) + \beta F_d(\beta) \right], & K < \mu \\ \frac{2\sigma\epsilon^3}{1+\epsilon^2} \left[\frac{d+\alpha^2}{d-1} f_d(\alpha) - \alpha(1 - F_d(\alpha)) \right], & K \geq \mu \end{cases} \quad (96)$$

with the notations

$$\begin{aligned} \alpha &= \frac{K - \mu}{\epsilon\sigma} \\ \beta &= \frac{\epsilon(K - \mu)}{\sigma}. \end{aligned}$$

The CMS swaplet is

$$V_S = \mu - K + 2\sigma(\epsilon - \frac{1}{\epsilon}) \frac{d}{d-1} f_d(0) \quad (97)$$

and the CMS floorlet is obtained by put-call parity as

$$V_F = V_C - V_S. \quad (98)$$

Proof. The spot price S follows this distribution under the T_P -forward measure such that the CMS caplet price is

$$V_C = E^{T_P} \left[(S_{T_F} - K)^+ \right] = \int_{-\infty}^{+\infty} (S - K)^+ f_{\mu,\sigma,\epsilon,d}(S) dS.$$

Let us define the parameter λ as

$$\begin{aligned} f_d(x) &= \lambda \left(1 + \frac{x^2}{d} \right)^{-\frac{d+1}{2}} \\ \lambda &= \frac{\Gamma(\frac{d+1}{2})}{\Gamma(\frac{d}{2}) \sqrt{d\pi}}. \end{aligned}$$

The price is then the integral

$$\begin{aligned} V_C &= \int_K^{+\infty} (S - K) f_{\mu,\sigma,\epsilon,d}(S) dS \\ &= \frac{2\lambda\epsilon}{\sigma(1+\epsilon^2)} \int_K^{+\infty} (S - K) \left[\left(1 + \frac{\beta^2(S)}{d} \right)^{-p} \mathbf{1}_{S < \mu} + \left(1 + \frac{\alpha^2(S)}{d} \right)^{-p} \mathbf{1}_{S \geq \mu} \right] dS \end{aligned} \quad (99)$$

with the definitions

$$\begin{aligned}\beta(S) &= \frac{\epsilon(S - \mu)}{\sigma} \\ \alpha(S) &= \frac{S - \mu}{\epsilon\sigma}.\end{aligned}\tag{100}$$

From this point we will use several times the identity

$$\int_a^b (X - c) \left(1 + \frac{X^2}{d}\right)^{-p} dX = \frac{1}{\lambda} \left[-\frac{d}{d-1} \left(f_d(b) \left(1 + \frac{b^2}{d}\right) - f_d(a) \left(1 + \frac{a^2}{d}\right) \right) - c \left(F_d(b) - F_d(a) \right) \right] \tag{101}$$

We separate the price in the cases $K \geq \mu$ and $K < \mu$ into

$$V_C = \frac{2\lambda\epsilon}{\sigma(1+\epsilon^2)} \left[\int_K^\mu (S - K) \left(1 + \frac{\beta^2(S)}{d}\right)^{-p} dS + \int_\mu^{+\infty} (S - K) \left(1 + \frac{\alpha^2(S)}{d}\right)^{-p} dS \right], \quad K < \mu \tag{102}$$

$$V_C = \frac{2\lambda\epsilon}{\sigma(1+\epsilon^2)} \int_K^{+\infty} (S - K) \left(1 + \frac{\alpha^2(S)}{d}\right)^{-p} dS, \quad K \geq \mu. \tag{103}$$

Using (101) it is then cumbersome but otherwise straightforward to calculate these integrals, finding the caplet prices as stated in (96).

The CMS forward in (97) is obtained by performing the similar integration

$$V_S = E^{T_P} [S - K] \tag{104}$$

and the CMS floorlet is obtained by put-call parity.

□

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